ER-to-Plasma Membrane Tethering Proteins Regulate Cell Signaling and ER Morphology

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SUMMARY

Endoplasmic reticulum-plasma membrane (ER-PM) junctions are conserved structures defined as regions of the ER that tightly associate with the plasma membrane. However, little is known about the mechanisms that tether these organelles together and why such connections are maintained. Using a quantitative proteomic approach, we identified three families of ER-PM tethering proteins in yeast: Ist2 (related to mammalian TMEM16 ion channels), the tricalbins (Tcb1/2/3, orthologs of the extended synaptotagmins), and Scs2 and Scs22 (vesicle-associated membrane protein-associated proteins). Loss of all six tethering proteins results in the separation of the ER from the PM and the accumulation of cytoplasmic ER. Importantly, we find that phosphoinositide signaling is misregulated at the PM, and the unfolded protein response is constitutively activated in the ER in cells lacking ER-PM tether proteins. These results reveal critical roles for ER-PM contacts in cell signaling, organelle morphology, and ER function.

INTRODUCTION

The endoplasmic reticulum (ER) is a vast membrane network of sheets and tubules with essential roles in protein modification and secretion, lipid synthesis, and calcium signaling. To execute its various functions, the ER is organized into distinct compartments, such as rough ER membranes studded with ribosomes and smooth ER enriched in lipid biosynthetic enzymes. In addition, the ER adopts diverse shapes and architectures in different organisms and specialized cell types (Shibata et al., 2006). For example, in cells dedicated to protein secretion, such as pancreatic acinar cells, the ER is an expansive system compactly arranged into stacks of membrane sheets. In contrast, in excitable cells, such as muscle cells, ER tubules form close associations with the plasma membrane (PM) that are important for calcium-inducible contractions (Carrasco and Meyer, 2011). Thus, cells have evolved multiple mechanisms for ER shape and architecture that support specialized functions performed in the ER.

The ER must coordinate each of its specialized functions with other organelles in the cell (e.g., the PM, Golgi, endosomes, lysosomes, and mitochondria). Vesicular trafficking between organelles has long been appreciated as a major mechanism for intracellular signaling and communication. However, recent studies have highlighted roles for membrane junctions between organelles—where two organelles are closely apposed without undergoing membrane fusion—as important sites for intracellular signaling (Carrasco and Meyer, 2011; Toulmay and Prinz, 2011). Although the ER is known to contact numerous organelles, ER-PM junctions are some of the most prominent. Since their initial discovery (Porter and Palade, 1957), ER-PM junctions have been observed in numerous organisms and cell types. One of the best-studied examples is the association of ER-localized stromal interaction molecule (STIM) proteins with the PM calcium channel Orai at ER-PM junctions in the store-operated calcium entry pathway (Liou et al., 2005; Park et al., 2009). Additional PM-ER junctions have been described in neurons, insect photoreceptor cells, and plants (Hayashi et al., 2008; Hepler et al., 1990; Suzuki and Hirosawa, 1994).

In yeast cells, large regions of the plasma membrane have an underlying network of cortical ER (cER) (Pichler et al., 2001; West et al., 2011). This network is associated with 20%–45% of the PM, with an average distance of 33 nm between the organelles (Pichler et al., 2001; West et al., 2011). Due to the close association of the PM and cER, ribosomes are excluded from the face of the cER adjacent to the PM (West et al., 2011), suggesting these structures have roles distinct from protein translocation into the ER. Consistent with this, a pathway has been described for the regulation of phosphoinositide (PI) lipid turnover at the PM by an ER-localized PI phosphatase (Stefan et al., 2011). In addition, the yeast *o*xy*s*terol-binding protein *h*omologs (Osh proteins) have been suggested to mediate sterol lipid transport at ER-PM contact sites (Schulz et al., 2009). Thus, junctions between the ER and the PM appear to be important sites for lipid metabolism and transport.

While ER-PM junctions have been known for several decades, our understanding of the mechanisms that tether the cortical ER to the PM and why such connections are established remains limited. The vesicle-associated membrane protein-associated protein (VAP) orthologs Scs2 and Scs22 have been implicated in PM-ER contact formation and cortical ER inheritance in yeast (Loewen et al., 2007). However, we know relatively little about the architecture and the structural components of ER-PM junctions. In this study, we have uncovered the mechanism of ER-PM tethering in yeast. Using quantitative proteomics approaches, we identified additional ER-PM tethering factors: Ist2 (a member of the TMEM16 ion channel family) and the tricalbin proteins (Tcb1, Tcb2, and Tcb3—orthologs of the extended synaptotagmin family). These proteins have been suggested to localize to the cortical ER, but functions for these proteins have remained unknown (Fischer et al., 2009; Toulmay and Prinz, 2011). We demonstrate that Ist2, the tricalbins, and the VAP orthologs Scs2/22 tether the cortical ER to the plasma membrane. Deletion of all six genes leads to a retraction of the ER away from the PM and redistribution of ER into internal structures. This collapse of cortical ER leads to elevated levels of a phosphoinositide lipid at the PM and an unexpected induction of the unfolded protein response (UPR) in the ER. Moreover, cells lacking cER have impaired growth under cellular stress conditions and require an intact UPR for viability.

RESULTS

Identification of Candidate PM-ER Tether Proteins

The cER and PM make extensive contacts, as observed in yeast cells by the close apposition of the ER protein Sec61-GFP and a PM marker (mCherry-2xPH$^{PLC\delta}$) (Figure 1A). At these sites, levels of the lipid phosphatidylinositol-4-phosphate (PI4P) in the PM are regulated in *trans* by the ER-localized PI phosphatase Sac1 (Stefan et al., 2011). However, the mechanisms that link the cER and PM (within 15 nm) and allow Sac1 to catalyze PI4P turnover at the PM are not well understood. The VAP orthologs Scs2/Scs22 have been implicated in PM-ER membrane tethering and Sac1-mediated PI4P turnover (Loewen et al., 2007; Stefan et al., 2011). Scs2 and its homolog Scs22 contain a single transmembrane domain and a major sperm protein (MSP) domain that binds FFAT motifs (two phenylalanines in an acidic tract) in lipid transfer proteins (Figure 1C; Loewen et al., 2003). However, cells lacking both Scs2 and Scs22 only accumulate a 2-fold increase in PI4P, while loss of Sac1 results in a nearly 10-fold increase (Foti et al., 2001; Stefan et al., 2011; Figure S4B available online). If Scs2/22 were the only cER-PM tethers, cells lacking these proteins should display a dramatic change in PI4P levels, similar to *sac1* mutants. Thus, additional tethering proteins that link the PM and ER likely exist.

Since Sac1 and Scs2 function at sites where the PM and ER are tightly apposed, we hypothesized that additional proteins involved in PM-ER tethering might interact with Sac1 and Scs2. We performed stable isotope labeling of amino acids in culture (SILAC) and identified Sac1 and Scs2-interacting proteins by quantitative mass spectrometry analysis (Figure 1B). We identified 26 and 32 high confidence interactions for Sac1 and Scs2, respectively (hits with greater than 10 peptides, a false positive rate <1%, and an Xpress Ratio above 20), and 17 proteins interacted with both Sac1 and Scs2, including each other (Tables S1 and S2). We also identified three reported Sac1-interacting proteins, Lcb1, Lcb2, and Dpm1 (Table S1; Breslow et al., 2010; Faulhammer et al., 2005) and two known Scs2-interacting proteins, Osh1 and Yet1 (Table S2; Loewen et al., 2003; Wilson et al., 2011). Many of the Sac1 and Scs2-interacting proteins have known functions in lipid metabolism, protein trafficking, and protein glycosylation, while only a small subset had no previously annotated function (Tables S1 and S2). Interestingly, we identified the reticulons, Rtn1 and Rtn2, and atlastin, Sey1, as Scs2-interacting proteins. These proteins have established roles in maintaining the structure of the ER (Hu et al., 2009; Voeltz et al., 2006). However, loss of these proteins does not disrupt PM-ER contacts (Hu et al., 2009; West et al., 2011).

Among the proteins identified, we considered three as candidate PM-ER tethers: Scs2, Ist2, and the yeast tricalbin proteins Tcb1 and Tcb3 (Figure 1B). Ist2, Tcb1, and Tcb3, along with the other tricalbin, Tcb2, were selected because they exhibit exclusively cortical ER localization (Toulmay and Prinz, 2012; Fischer et al., 2009; Figures 1D and S1B) and have no annotated functions. Ist2 and the tricalbins also have protein architectures that might be expected for an organelle-tethering protein: transmembrane domains and cytoplasmic lipid-binding domains. Ist2 is comprised of eight transmembrane domains and a long cytoplasmic carboxyl-terminus that binds lipids and localizes to the cell cortex (Fischer et al., 2009; Figure 1C). The tricalbins contain transmembrane domains in their N-termini and lipid-binding C2 domains in their long cytoplasmic carboxyl-termini (Figure 1C, Schulz and Creutz, 2004). The tricalbins also possess a synaptotagmin-like mitochondrial lipid-binding protein (SMP) domain that is found in other proteins localized to ER-organelle contact sites and is homologous to a domain found in lipid transfer proteins (Kopec et al., 2010; Toulmay and Prinz, 2012).

We next confirmed the localization of Ist2 and Tcb3 to the cER. At the midsection of yeast cells, GFP-tagged Ist2 and Tcb3 overlapped with the ER marker DsRed-HDEL, specifically at the cortical ER (Figure 1D). Significant colocalization of both Ist2 and Tcb3 was observed with the ER marker at the periphery of these cells (Figure 1D). Notably, regions devoid of ER lacked GFP signal. The other tricalbin family members, Tcb1 and Tcb2, also localize to the cER (Figure S1B; Toulmay and Prinz, 2012). However, both Ist2 and the tricalbins have been suggested to be PM proteins (Creutz et al., 2004; Jüschke et al., 2005). To resolve these differing results, we performed equilibrium density gradient fractionations that efficiently separate ER and PM membranes. Consistent with our microscopy results that localized Ist2 and Tcb3 in the cER, both 3xHA-Ist2 and Tcb3-3xHA were present in fractions containing the ER markers Sac1-3xFLAG and Dpm1 and absent from the PM marker Pma1 fraction (Figure 1E).

Deletion of Tethers Results in a Loss of cER and Accumulation of Large Cytoplasmic ER Structures

To determine if Ist2 and the tricalbin proteins function by tethering the cER to the PM, we constructed multiple deletion mutants of each of the tether candidates and scored Sec61-GFP localization as an ER marker. In wild-type cells, the ER is organized into nuclear ER, a few cytoplasmic ER tubes and sheets, and cortical structures juxtaposed to the periphery of the cells (Figure 2A). Deletion of all six genes (*ist2, scs2/22,* and *tcb1/2/3*) resulted in a striking phenotype-substantial reduction in cER, as large areas of the cell periphery have no associated ER (differential interference contrast/GFP merge) (Figure 2A). Occasionally, small regions of the ER closely apposed to the cell cortex were observed (Figure 2A, arrow). In addition, collapsed ER structures accumulated in the cytoplasm around the nucleus and in the emerging bud of mutant cells (Figure 2A). ER inheritance appeared to be normal in cells lacking the six genes, as nonnuclear ER was present in the bud before nuclear migration (Figure S2, asterisk). To verify our results with Sec61-GFP, we examined the morphology of the ER using another

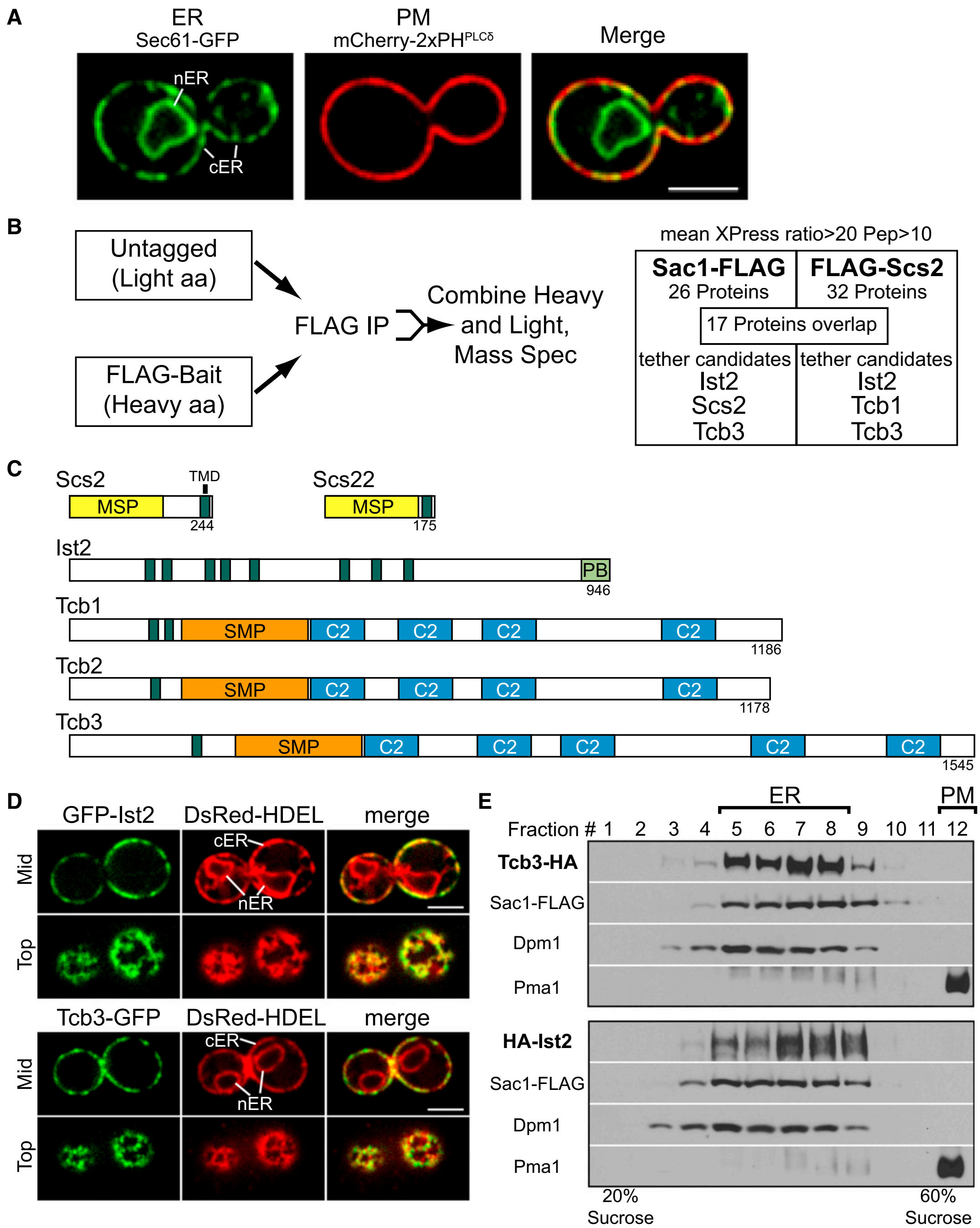

Figure 1. Identification of Candidate PM-ER Tethers

(A) ER marker Sec61-GFP (green) and PM marker mCherry-2xPH$^{PLC\delta}$ (red) expressed in wild-type cells. Cortical ER and nuclear ER are labeled. Scale bar, 3 μm.

(B) Schematic of Sac1 and Scs2 SILAC experiments followed by quantitative mass spectrometry. See also Tables S1 and S2.

(C) Diagram of candidate tethering proteins. Amino acid length is indicated. MSP, major sperm protein domain; TMD, transmembrane domain; PB, polybasic domain; SMP, synaptotagmin-like-mitochondrial lipid-binding protein.

(D) GFP-Ist2 and Tcb3-GFP localize to the cER in wild-type cells (ER is marked by DsRed-HDEL). Nuclear ER and cortical ER are indicated. Scale bar, 3 μm. See also Figure S1.

(E) Equilibrium density fractionation of cells expressing 3xHA-Ist2, Tcb3-3xHA, and Sac1-3xFLAG. Fractions were collected and analyzed by immunoblotting for Sac1-3xFLAG (ER), Dpm1(ER), and Pma1(PM) as organelle markers.

marker, GFP-HDEL, in multiple sections along the Z-axis in cells. In wild-type cells, the ER network was clearly visible at the cell cortex and remained associated with the periphery through all Z sections observed (Figure 2B). However, in cells lacking the tether proteins, the ER was no longer closely apposed to the PM, as GFP-HDEL was not observed at the periphery and mainly

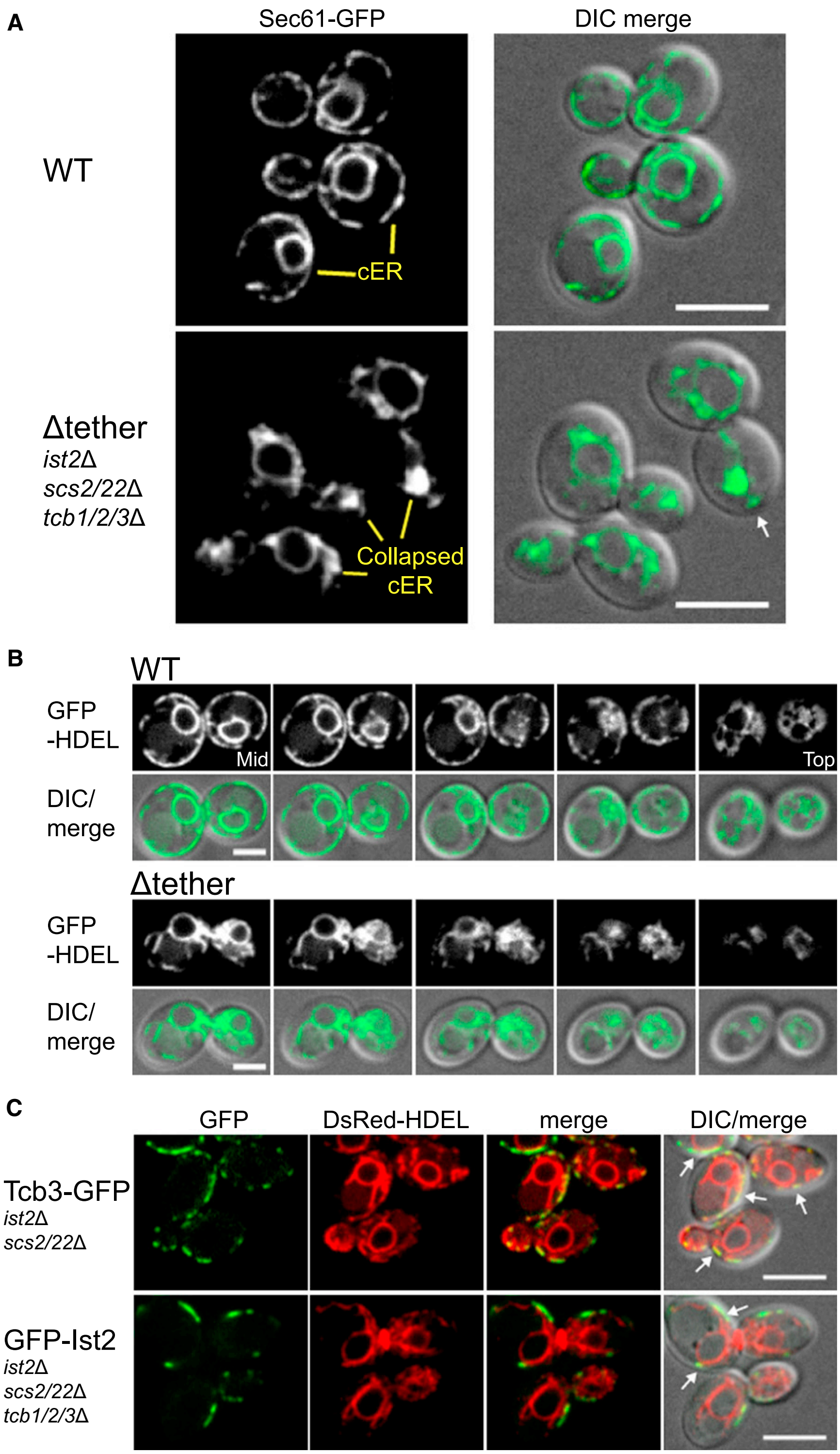

Figure 2. Scs2/22, Ist2, and the Tricalbins Tether the cER to the PM in Yeast

(A) Localization of the ER marker Sec61-GFP in wild-type and Δtether cells (*ist2*Δ, *scs2/22*Δ, and *tcb1/2/3*Δ). cER is indicated in wild-type cells and collapsed ER in Δtether cells. Arrow indicates potential remaining cER in Δtether cell. Scale bar, 5 μm. See also Figure S2.

(B) Z stacks of wild-type and Δtether cells expressing the ER marker GFP-HDEL. Each step is 0.4 μm. Scale bar, 2.5 μm.

(C) Tcb3-GFP in *ist2*Δ, *scs2/22*Δ cells and GFP-Ist2 localization in *scs2/22*Δ, *tcb1/2/3*Δ cells expressing DsRed-HDEL. Arrows show regions labeled by the ER marker and Ist2 or Tcb3. Scale bar, 5 μm.

localized internally (Figure 2B). Based on these results, we termed the strain lacking all six tethering proteins as Δtether cells and use this name throughout the study.

We observed a range of effects upon loss of the VAP proteins, Ist2, or tricalbins either alone or in combination. In *ist2* single mutant cells and *tcb1/2/3* triple mutant cells, there was no significant loss of cER (Figure S2, marked by yellow arrows). Likewise, cER structures were readily observed in *ist2 tcb1/2/3* quadruple mutant cells, suggesting that the Scs2 and Scs22 VAP proteins were sufficient to form cER (Figure S2). In *scs2 scs22* mutant cells, cER structures were reduced compared to wild-type cells—but not absent—consistent with roles for Ist2 and the tricalbins in cER formation (Figure S2). Accordingly, upon loss of Ist2 or the tricalbins in *scs2/22* cells (*ist2 scs2/22* and *scs2/scs22 tcb1/2/3* cells) (Figure S2), we observed a further decrease in cER and an increase in enlarged cytoplasmic ER structures (Figure S2, red arrows) compared to *scs2/22* cells. We next confirmed that Ist2 and the tricalbins are sufficient for cER formation and localize to the remaining cER in cells expressing only one type of tether. In cells lacking all other tethers, both GFP-Ist2 (in *tcb1/2/3, scs2/22* cells) and Tcb3-GFP (in *ist2, scs2/22* cells) localized to the regions of remaining cortical ER, suggesting that the cortical ER that is still present in an *scs2/22* mutant is tethered by Ist2 and the tricalbins (Figure 2C). These results also demonstrated that Ist2 and the tricalbins do not require the other tethers for their localization to PM-ER contacts or tethering function.

Ultrastructural Analysis of cER Mutants

Because we observed cytoplasmic ER accumulations in the Δtether mutant cells by fluorescence microscopy, we next examined the morphology of these structures at higher resolution using electron microscopy. In wild-type cells, ER membranes appeared as dark stained structures at the periphery of the cell (cER) and as a discontinuous ring within the cell (nuclear ER) (Figure 3A, labeled in blue). The cER (ER adjacent to the PM, labeled in purple) is the

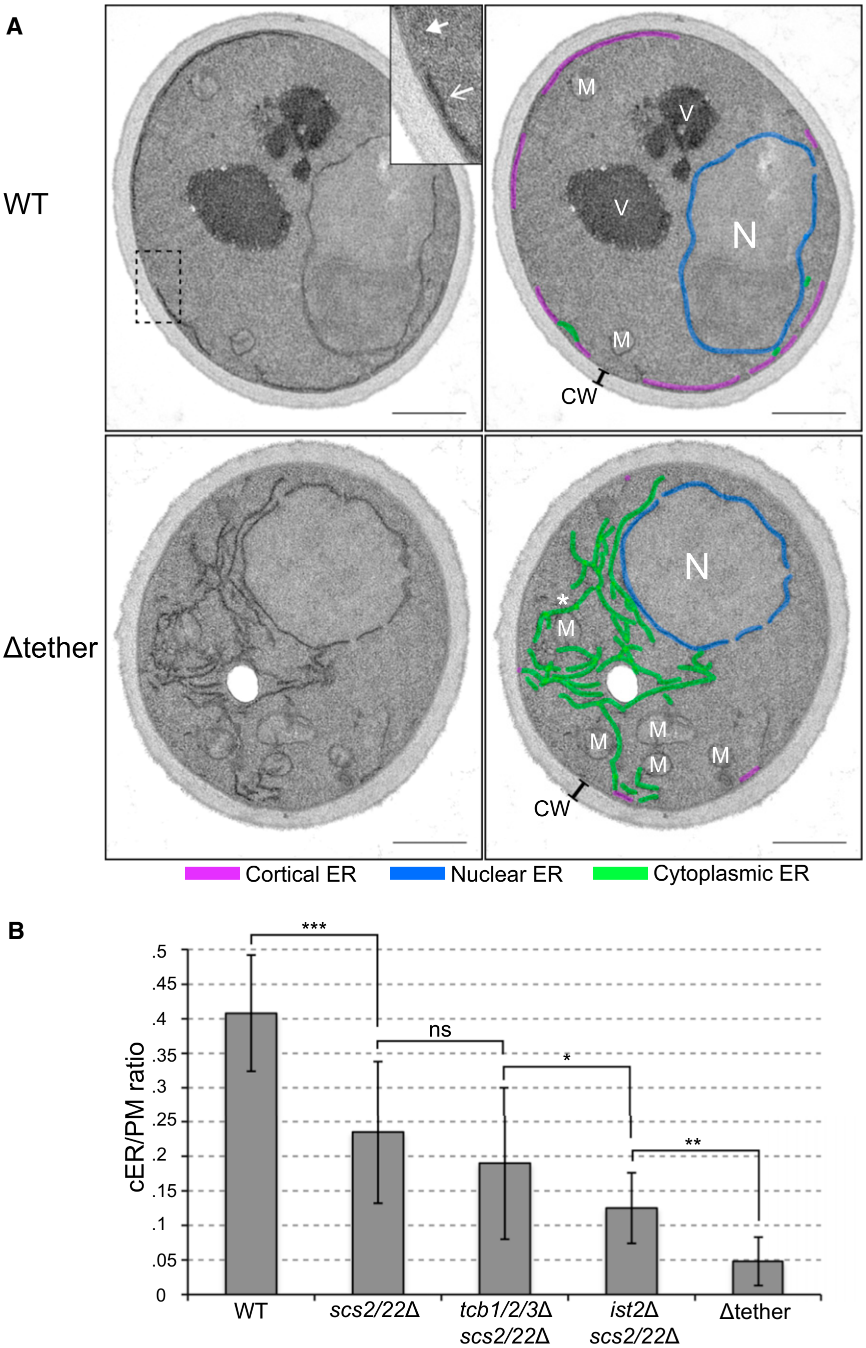

Figure 3. Morphology of Δtether Cells and Quantification of cER in Tether Mutants

(A) Electron microscopy of wild-type and Δtether cells. Inset of wild-type cell is an enlarged region showing PM associated with (open arrow) and without ER (closed arrow). Different ER structures are labeled in the right panels: cER in purple, nuclear ER in blue, and cytoplasmic ER in green. Mitochondria (M), nucleus (N), cell wall (CW), and vacuole (V) are also labeled. A mitochondrial-ER contact is shown by an asterisk. Scale bar, 500 nm. See also Figures S3A and S3B.

(B) Quantification of cER expressed as a ratio of the length of cER/length of the PM in wild-type and mutant cells. Error bars show SD (n = 30 for each strain); significance was determined by one-way ANOVA with postprocessing to correct for multiple comparisons (***$p < 0.001$; **$p < 0.01$; *$p < 0.05$; ns, not significant; $p > 0.05$). See also Figure S3C.

most abundant nonnuclear ER in wild-type cells, as very little cytoplasmic ER (green) was observed. In striking contrast to the ER morphology in wild-type cells, the majority of the ER in Δtether mutants is not at the periphery. Instead, retracted masses of ER accumulated within the cytoplasm and likely account for the collapsed ER structures observed by fluorescence microscopy (Figure 3A). In addition to the cytoplasmic network, some Δtether cells also contained what appeared to be long ER tubules (Figure S3A). However, these structures are more likely extended ER sheets, as a tubule would not remain in the same plane of a thin EM section (~70 nm) for such great distances (up to 5–7 μm). Additional examples of Δtether mutant cells with collapsed ER structures and long cytoplasmic ER structures are shown in Figure S3B. Despite the large decrease in ER-PM contacts, mitochondria-ER junctions appear intact (Figure 3A).

Next, we quantified the loss of cER structures upon sequential deletion of the ER tether proteins. We decided to focus on mutants that had obvious cER morphology defects by fluorescent microscopy, and thus we used *scs2/22* double mutant cells as a starting point. To quantitate the loss of cER in the mutants, we used a ratio of the sum length of cER segments (purple) over the circumference of the PM. Consistent with published results (West et al., 2011), in wild-type cells, approximately 40% of the PM was associated with cER structures (0.4 cER/PM ratio) (Figure 3B). In *scs2/22* mutants, the cER/PM ratio was decreased by nearly one-half (24% of the PM associated with cER) (Figures 3B and S3C), consistent with previously published observations (Loewen et al., 2007). Upon further loss of Ist2, the cER/PM ratio decreased an additional 2-fold (13% of the PM associated with cER structures in *ist2, scs2/22* triple mutant cells) (Figures 3B and S3C). Upon deletion of all six genes (Δtether mutant cells), the cER/PM ratio decreased greater than 8-fold, as compared to wild-type cells (only 4.8% of the PM associated with cER) (Figure 3B). Surprisingly, the cER/PM ratio for *tcb1/2/3*, *scs2/22* mutant cells was not significantly different than *scs2/22* mutants alone, and loss of the tricalbins only exhibited an effect when the other tethers (Scs2/22 and Ist2) were absent (compare *ist2 scs2 scs22* cells to Δtether cells) (Figures 3B and S3C). Our results using both fluorescence and electron

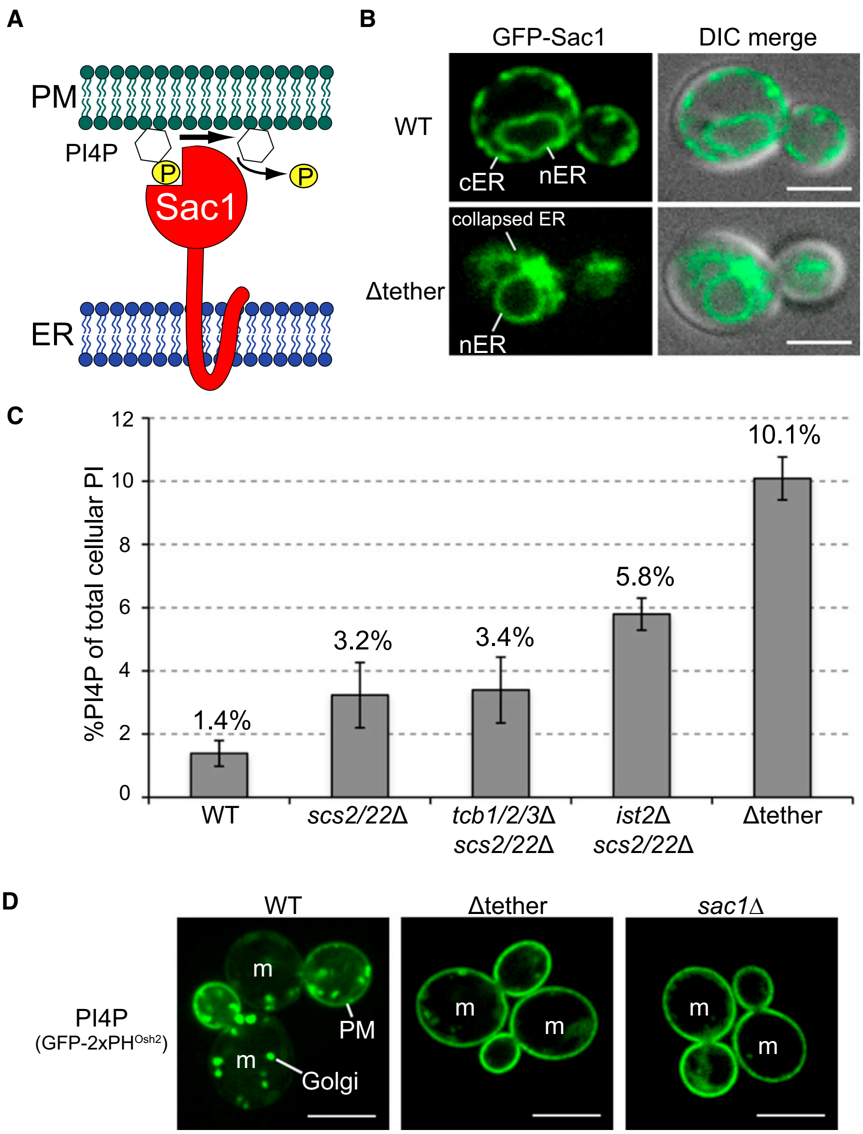

Figure 4. Loss of ER-PM Contacts Results in Misregulation of PI4P at the PM

(A) Schematic for Sac1 phosphatase function. Sac1 is an integral ER membrane protein that dephosphorylates the lipid PI4P at the PM.

(B) Localization of GFP-Sac1 in wild-type and Δtether cells. Scale bar, 3 μm. See also Figure S4A.

(C) Cellular PI4P levels in wild-type and ER tether mutant cells, as measured by ^{3}H-inositol labeling and HPLC analysis. Error bars show SD (n = 3). See also Figure S4B and Table S3.

(D) Localization of the PI4P reporter GFP-2xPH^{Osh2} in wild-type, Δtether, and *sac1*Δ cells. PM and Golgi pools of PI4P are indicated. Mother cells are labeled (m). Scale bar, 5 μm.

microscopy suggest a hierarchy among the PM-ER tethers, where Scs2/22 provide a significant contribution, followed by Ist2, and then the tricalbins. It should be noted that, in Δtether mutant cells, even though there is a large redistribution of cER to the cytoplasm, there appears to be small regions of cER present, suggesting the presence of additional unknown tethering proteins (Figures 3 and S3B).

PI4P Accumulates at the PM in Δtether Mutant Cells

The PI phosphatase Sac1 dephosphorylates PI4P on the PM in *trans* from the ER (Figure 4A; Stefan et al., 2011). In cells where PM-ER connections are disrupted, Sac1 should have less access to its substrate at the PM and PI4P levels should increase. We first confirmed that Sac1 was mislocalized to intracellular ER structures in Δtether cells. Similar to other ER markers, GFP-Sac1 accumulated in the collapsed ER in Δtether cells (Figure 4B). Sac1 expression levels were not significantly different in whole cell lysates from wild-type and Δtether mutant cells (Figure S4A). To monitor Sac1 function, we performed ^{3}H-inositol-labeling experiments and high-performance liquid chromatography (HPLC) analysis to measure cellular PI levels in wild-type and the tether mutant strains (Table S3). Consistent with a requirement for ER-localized Sac1 to be in close proximity to the PM, mutants with reduced cER displayed increased PI4P levels (Figure 4C). Increases in PI4P correlated with decreases in cER, as *scs2/22* and *scs2/22, tcb1/2/3* mutant cells had a modest increase in PI4P (greater than 2-fold), while *ist2*, *scs2/22* mutants with significantly reduced cER, accumulated higher PI4P levels (4-fold). In cells lacking only Ist2 (*ist2* mutant cells) or the tricalbins (*tcb1/2/3* cells), PI4P levels were similar to wild-type (Table S3). In Δtether cells, in which the cER/PM ratio is reduced more than 8-fold (only 5% of the PM is associated with cER), PI4P levels increased greater than 7-fold (Figure 4C).

In addition to PI4P at the PM, the Sac1 PI phosphatase regulates pools of PI4P at the Golgi and ER (Faulhammer et al., 2005). We then addressed whether the ER tether proteins specifically regulate PI4P levels at the PM. For this, we visualized PI4P in wild-type and Δtether cells using a fluorescent reporter that binds to PI4P (GFP-2xPH^{Osh2}; Roy and Levine, 2004). In wild-type cells, the PI4P sensor was present at both Golgi compartments and the PM, with enriched PM localization in the bud (Figure 4D). However, similar to *sac1* mutant cells, the PI4P reporter was shifted to the PM in Δtether cells, with an even distribution in both mother and daughter cells (Figure 4D).

If the ER tether proteins regulate PI4P metabolism through the Sac1 phosphatase, then loss of ER tether proteins should not result in a significant increase in PI4P in cells lacking Sac1 function. We measured PI4P levels in an *ist2*, *scs2/22*, *sac1*^{ts} temperature conditional strain (*ist2*, *scs2/22*, *sac1* quadruple null mutants were inviable; our unpublished data) and *sac1*^{ts} single mutant cells at the nonpermissive temperature. PI4P

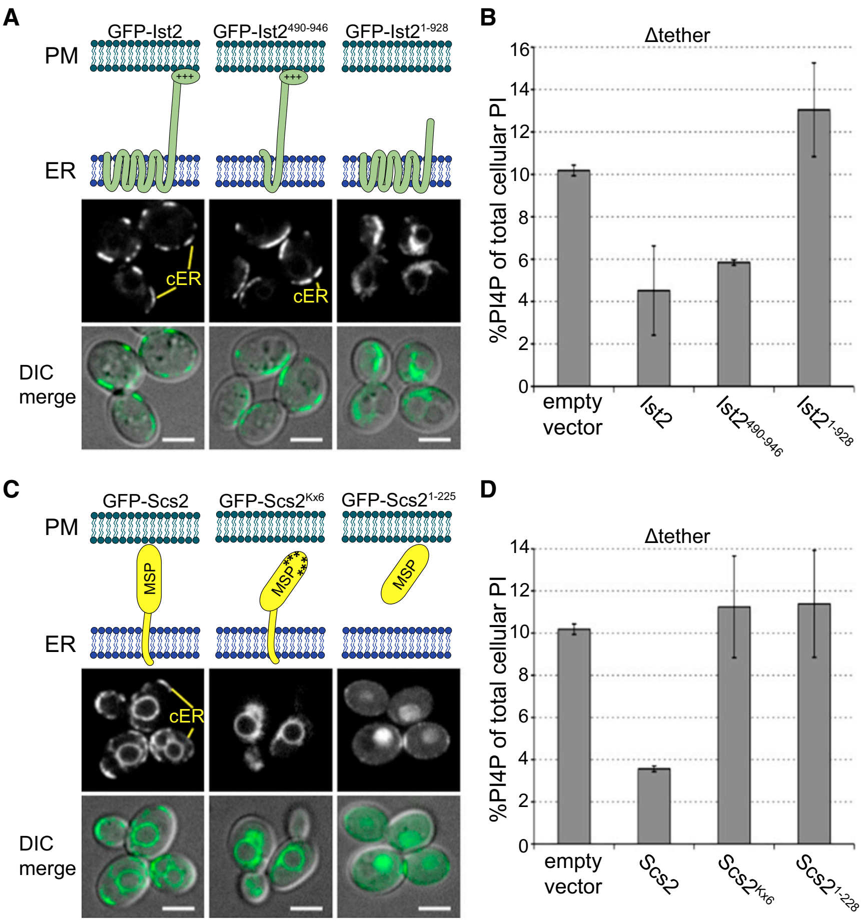

Figure 5. Ist2 C Terminus Is Required for Tethering and Association with the PM

(A) Localization and models of GFP-Ist2, GFP-Ist2$^{490–946}$, and GFP-Ist2$^{1–928}$ constructs expressed in Δtether cell. Scale bar, 3 μm. See also Figures S5A–S5D.

(B) PI4P levels in Δtether cells expressing GFP-Ist2, GFP-Ist2$^{490–946}$, or GFP-Ist2$^{1–928}$. Error bars show SD (n = 2). See Table S3.

(C) Localization of GFP-Scs2, GFP-Scs2^{Kx6}, and GFP-Scs2$^{1–225}$ in Δtether cells. Scale bar, 3 μm. See also Table S4 and Figures S5E and S6A.

(D) PI4P measurements in Δtether cells carrying empty vector or expressing *SCS2*, *scs2^{Kx6}*, or *scs2$^{1–225}$*. Error bars show SD (n = 2). See also Table S3.

levels in *ist2*, *scs2/22*, *sac1^{ts}* quadruple mutant cells were not significantly increased as compared to PI4P in *sac1^{ts}* single mutant cells (13.4% ± 1% and 11.6% ± 2.5%, respectively) (Figure S4B). Similar to our previous results, PI4P levels were increased (>3-fold) in *ist2*, *scs2/22* triple mutant cells as compared to wild-type cells (Figure S4B). Taken together, our results indicate that the ER tether proteins facilitate Sac1 interactions with the PM to regulate PI4P.

Identification of Regions in Scs2, Ist2, and Tcb3 Involved in ER-PM Tethering and Localization

To better understand the mechanisms of ER-PM tethering, we identified domains in the ER tether proteins necessary for their function. We focused on Scs2 and Ist2, since our results implicated these proteins as key ER-PM tethers. Ist2 consists of eight transmembrane domains comprising a putative ion channel followed by a long cytoplasmic C terminus containing a polybasic region rich in lysine and histidine residues necessary for cortical localization of Ist2 (Jüschke et al., 2005). The Ist2 polybasic region also binds to PI(4,5)P_2-containing liposomes in vitro (Fischer et al., 2009). Similar to our previous results, expression of full-length Ist2 in Δtether cells was sufficient to restore some cER (Figures 2C, 5A, and S5D). Interestingly, the last two transmembrane domains and the extended C terminus (GFP-Ist2$^{490-946}$) were sufficient to form cER when expressed in Δtether cells (Figures 5A and S5D). The C-terminal polybasic domain was required for cER tethering, as the C-terminal truncated protein (GFP-Ist2$^{1-928}$) did not restore cER and instead localized to cytoplasmic ER structures in Δtether cells (Figures 5A and S5D). As an independent test for tethering function, we measured PI4P levels in Δtether mutant cells expressing the Ist2 constructs. Consistent with the ability to form cER, wild-type GFP-Ist2 and the mutant protein consisting of the last two transmembrane domains and the C terminus (GFP-Ist2$^{490-946}$) were able to reduce PI4P levels, but Ist2 lacking the polybasic domain (GFP-Ist2$^{1-928}$) was not (Figure 5B). Thus, the C terminus was necessary for Ist2 tethering and the proposed ion channel of Ist2 was dispensable for ER-PM tethering.

Since the C-terminal polybasic domain was necessary for Ist2 to tether the ER and PM, we tested whether the polybasic domain was sufficient for PM targeting when expressed on its own. A previous study suggested it has weak, if any, affinity for the PM (Jüschke et al., 2005). To increase the binding affinity for its target at the PM, we generated a tandem polybasic domain GFP construct (GFP-2xIst2$^{928–946}$). Strikingly, the tandem Ist2 polybasic domain (GFP-2xIst2$^{928–946}$) tightly associated with the PM (Figure S5A) compared to a single polybasic domain (Figure S5B) and still localized to the PM in Δtether cells (Figure S5C).

The VAP protein Scs2 contains a single transmembrane domain (TMD) and a cytoplasmic MSP domain (Figures 1C and 5C). As expected, expression of full-length Scs2 in Δtether cells was sufficient to restore cER (Figures 5C and S5E) and lower cellular PI4P levels (Figure 5D). A truncated form of Scs2 lacking the TMD (GFP-Scs2$^{1–225}$) did not localize to the ER and instead was in the nucleus and weakly at the PM at sites of polarized growth (Figures 5C and S5E; Loewen et al., 2007). The mutant Scs2 protein lacking its TMD failed to reduce PI4P levels in the Δtether mutant cells (Figure 5D), suggesting Scs2 must be anchored in the ER to function as a tether. A mutant form of Scs2 (Kx6) bearing substitutions in basic lysine residues within the MSP domain involved in binding to FFAT peptides and

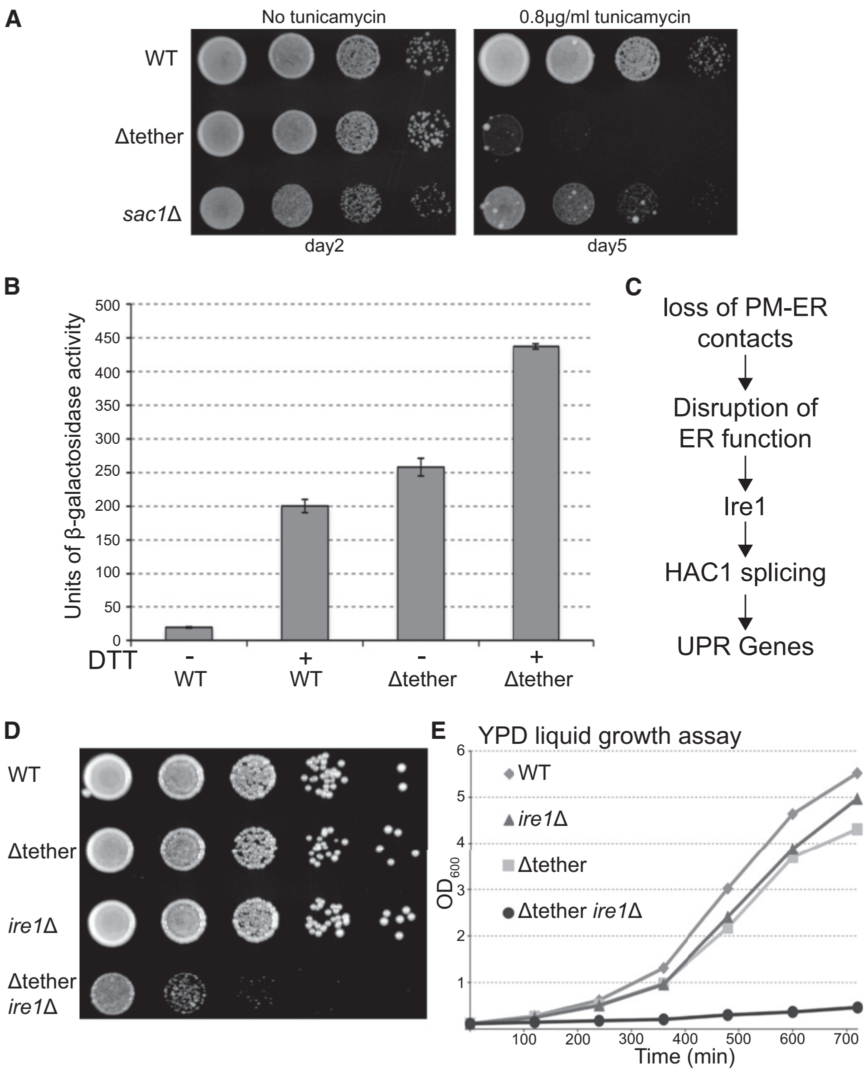

Figure 6. The UPR Is Critical for the Growth of Cells Lacking PM-ER Contacts

(A) Serial dilutions of wild-type, Δtether, and *sac1*Δ cells grown for indicated times at 26°C on yeast nitrogen base (YNB) or YNB + 0.8 μg/ml tunicamycin.

(B) UPR assay using UPRE-LacZ reporter. Induction of the UPR was measured by β-galactosidase activity in indicated strains treated with or without 8 mM DTT for 1 hr. Error bars show SD (n = 3 assays performed in duplicate).

(C) Diagram of the UPR pathway and a potential link to disruption of PM-ER contacts. Loss of PM-ER contacts results in ER dysfunction that is sensed by Ire1, which induces the UPR.

(D) Serial dilutions of wild-type, Δtether, *ire1*Δ, and Δtether /*ire1*Δ mutant cells grown on YPD plates at 26°C.

(E) Liquid growth assays of wild-type, Δtether, *ire1*Δ, and Δtether/*ire1*Δ mutant cells. Cells were grown in YPD at 26°C and OD_{600} measurements were taken at indicated time points.

See also Figure S7.

potentially to PI lipids (Kagiwada and Hashimoto, 2007) failed to restore cER (Figures 5C and S5E) or lower PI4P levels in Δtether mutant cells (Figure 5D). In addition to the Kx6 mutant, we also tested several other substitutions within the MSP domain of Scs2 (Table S4). The Scs2 mutant proteins showed a wide range of activity, including full wild-type function (e.g., L86A), partial activity (e.g., K84A, L86A), and complete loss of function (e.g., the T41A, T42A and P44S, P51S mutant proteins) (Table S4). We also tested whether the MSP domain was sufficient for ER-PM tethering when fused to another ER membrane protein. Expression of a GFP-MSP-Sac1 fusion protein in Δtether cells was sufficient to restore cER (Figure S6A). Taken together, these results implicate the MSP domain in Scs2 tethering function, consistent with previous findings (Loewen et al., 2007).

Our results showed that the tricalbins are sufficient for cER formation and that Tcb3-GFP remained localized to ER-PM contacts in cells lacking Ist2 and Scs2/22 (Figure 2C). The tricalbins contain multiple lipid-binding C2 domains (Figure 1C) that could be involved in ER-PM targeting. Interestingly, a mutant form of Tcb3 lacking the C2 domains (Tcb3$^{1–419}$-GFP) was observed in both nuclear and cortical ER structures, in contrast to full-length Tcb3-GFP that localized exclusively to the peripheral ER (Figure S6B). Thus, the C2 domains may serve in stabilizing the tricalbin proteins at ER-PM junctions, consistent with previous findings (Toulmay and Prinz, 2012).

Cells Lacking PM-ER Contacts Require the UPR for Survival

Due to the dramatic changes in ER morphology, we reasoned that ER functions might be affected in the Δtether cells. We first examined if there was a defect in protein trafficking from the ER. However, glycosylation and delivery of the cargo protein carboxypeptidase Y (CPY) to the vacuole appeared normal, implying that ER export is unaffected by the change in ER morphology in Δtether cells (Figure S7A). We then tested whether the Δtether cells were sensitive to ER stress induced by tunicamycin. Tunicamycin blocks N-linked protein glycosylation in the ER leading to misfolding of both luminal and integral membrane proteins. Interestingly, the Δtether strain was hypersensitive to tunicamycin-induced ER stress as compared to wild-type cells and *sac1* mutant cells (Figure 6A). Thus, the growth defect of Δtether cells under ER stress conditions was not simply due to loss of Sac1 function.

The sensitivity of Δtether cells to an ER stress suggested a potential impairment of the UPR. The UPR detects misfolded proteins in the ER and relays this signal to the nucleus, resulting in increased expression of ER chaperones, lipid synthesis enzymes, ER-associated degradation machinery (ERAD), and membrane-trafficking protein (Travers et al., 2000). To determine if ER-PM contacts are required for the UPR, we measured induction of a UPRE-LacZ reporter in wild-type and Δtether cells (Cox and Walter, 1996). Untreated wild-type cells displayed low levels of UPRE-LacZ activity under basal conditions, but UPRE-LacZ activity was increased approximately 10-fold when wild-type cells were exposed to an ER stress, the reducing agent dithiothreitol (DTT) (Figure 6B). Unexpectedly, Δtether cells displayed constitutive UPR signaling, as there was a surprisingly high level of UPRE-LacZ activity (comparable to DTT-induced levels in wild-type cells), even without the addition of DTT. The UPRE-LacZ activity further increased when Δtether cells were treated with DTT (Figure 6B). To determine if the constitutive induction of the UPR correlated with defects in cER formation, we measured basal UPR activity in each of the tether mutants. Interestingly, only mutants with cER defects constitutively induced the UPR and basal activity increased with decreased cER formation (Figure S7B). The Δtether cells displayed significantly higher basal UPR signaling than *sac1* mutants (Figure S7B), implying ER stress in the Δtether cells was not entirely due to loss of Sac1 function. As an independent test, basal expression of Kar2 (yeast BIP, a protein induced during UPR) was moderately elevated in Δtether cells as compared to wild-type cells in the absence of ER stress (Figure S7C). Both wild-type and Δtether cells increased Kar2 expression in response to tunicamycin, indicating that Δtether cells possess an intact UPR signaling pathway (Figure S7C).

The Δtether cells undergo continuous UPR signaling, even in the absence of exogenous ER stress-inducing agents (e.g., tunicamycin or DTT), implying that the loss of PM-ER contacts disrupts one or more processes in the ER. While Δtether cells grew normally under nonstress conditions, Δtether cells were hypersensitive to tunicamycin, as if they cannot overcome additional ER stress. These results suggested that the change in ER morphology and increased ER stress experienced by Δtether cells is compensated by the UPR (Figures 6C). We then predicted that loss of UPR signaling would compromise the viability of Δtether cells. To test if the UPR was critical, we deleted the *IRE1* gene that encodes kinase and ribonuclease activities required for UPR signaling. Compared to wild-type, Δtether, and *ire1* cells, the growth of *ire1* Δtether mutant cells was severely impaired in both serial dilution plating assays and liquid cultures (Figures 6D and 6E). Loss of Hac1, a transcription factor downstream of Ire1 (Cox and Walter, 1996), also resulted in a severe synthetic growth defect in Δtether mutant cells (Figure S7D). The synthetic relationship between loss of PM-ER contacts and the UPR was not due to impaired protein trafficking out of the ER, as CPY was efficiently sorted in the *ire1* Δtether mutant (Figure S7E). Thus, the Δtether cells are defective for critical ER functions that are compensated by upregulation of the UPR.

DISCUSSION

Multiple Conserved Tethering Proteins Form ER-PM Junctions

PM-ER membrane contact sites are major sites for calcium and lipid signaling in eukaryotic cells. Yet relatively little is known about the structural components of membrane junctions between these organelles. We have identified Ist2 (related to the mammalian TMEM16 family of ion channels) and the tricalbins (C2 domain-containing proteins similar to the extended synaptotagmin-like proteins E-Syt1/2/3) as ER tether proteins that

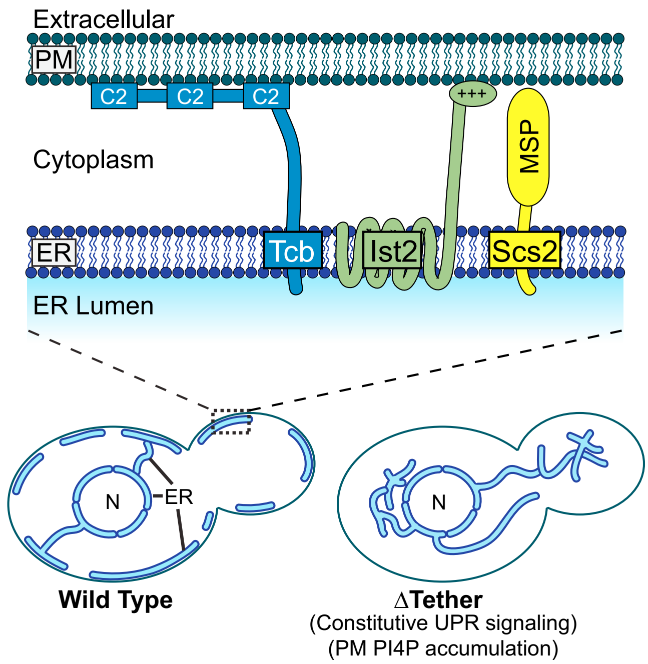

Figure 7. Model for PM-ER Tethering in Yeast
Three families of integral ER proteins tether the cortical ER to the PM: the VAP proteins Scs2/22, Ist2, and the tricalbins. Loss of ER-PM tethers results in dramatic changes in ER morphology, elevated PI4P levels at the PM, and activation of the UPR.

function with the VAP orthologs Scs2 and Scs22 to establish junctions between the ER and PM. Upon loss of all six ER tether proteins, the morphology of the cortical ER was drastically altered from a network of sheets and tubes attached to the PM in wild-type cells to a near total loss of PM-ER contacts and accumulations of ER in the cytoplasm. Importantly, our results suggest critical roles for PM-ER contacts in regulating phosphoinositide signaling at the PM and ER morphology and homeostasis (Figure 7).

Of the PM-ER tethering proteins analyzed in our study, Scs2 appeared to be the most important. Loss of Scs2 alone resulted in a clear reduction in cER compartments (Loewen et al., 2007; Figure S2A). Though Scs2 has a critical role in PM-ER tethering, Scs2 and other VAP family members may function as general tethers for several ER-organelle connections. VAP proteins have been implicated in ER-Golgi and ER-endosome contacts in mammalian cells (Peretti et al., 2008; Rocha et al., 2009). Scs2 is not enriched at specific ER-organelle contact sites and instead localizes throughout the ER (Figure S1A). These results suggest that the VAP proteins may cycle between at least two states: an active tethering state at organelle junctions and an inactive state diffuse throughout the ER network. This idea is consistent with VAP localization and may allow the VAP proteins to make contacts with several organelles as needed through regulated interactions with additional tethering factors. How the VAP protein Scs2 is regulated and the mechanisms by which Scs2 facilitates ER tethering are unclear. One possibility is that Scs2 binds to specific FFAT-containing *o*xysterol-binding protein *r*elated *p*roteins (ORP) family members at membrane contact sites. Several ORP family members have been observed at ER-organelle contact sites (Peretti et al., 2008; Rocha et al., 2009; Schulz et al., 2009; Stefan et al., 2011). However, ORPs (e.g., the Osh proteins in yeast) do not appear to be required for PM-ER tethering in vivo (Loewen et al., 2007; Schulz et al., 2009; Stefan et al., 2011). The Scs2 MSP domain has been suggested to bind negatively charged lipids, such as PI4P (Kagiwada and Hashimoto, 2007). However, PI4P is not sufficient for Scs2 localization, as the MSP domain targets to sites of polarized growth, and it was not observed at Golgi compartments that are enriched in PI4P (Loewen et al., 2007; Figure 5). Thus, Scs2 may interact with as yet unidentified proteins or lipids on target organelles to facilitate tethering.

Unlike the VAP orthologs Scs2/22, Ist2 and the tricalbins exclusively localized to the cortical ER, where they likely function as specific tethers between the cER and PM (Figure 1D; Fischer et al., 2009; Toulmay and Prinz, 2012). Their restricted localization to the cER implies that both Ist2 and the tricalbins make extensive contacts with the PM. The tricalbins may bind directly to PM lipids, as each contains lipid-binding C2 and SMP domains (Schulz and Creutz, 2004; Toulmay and Prinz, 2012). The C terminus of Ist2 bound to PI(4,5)P_2 containing liposomes in vitro and was both necessary and sufficient for PM interaction (Figure 5; Fischer et al., 2009). Interestingly, a recent study found that Ist2 regulates the close association of cER with the PM (within 30 nm) (Wolf et al., 2012), consistent with the idea that Ist2 directly contacts the PM.

Similar to Scs2, Ist2 and the tricalbins are conserved in higher eukaryotes, including mammals. Ist2 is related to the TMEM16/anoctamin proteins, some of which are calcium-activated chloride channels (Caputo et al., 2008; Yang et al., 2008). Interestingly, TMEM16a contains a cytoplasmic region that binds PI(4,5)P_2 in vitro and targets to cER-PM junctions in yeast cells (Fischer et al., 2009). The tricalbins are related to the mammalian extended synaptotagmin proteins (E-Syt1/2/3), which share a similar architecture of N-terminal membrane-anchoring domains followed by cytoplasmic SMP and C2 domains. Although some PM-ER tethers in higher eukaryotes are known (e.g., STIM proteins and junctophilin), we propose Ist2 and the tricalbin orthologs have conserved roles as ER-organelle tethers in other eukaryotes.

ER Inheritance and Establishment of the Cortical ER in Daughter Cells

It is worth noting that Ist2, Tcb2, and Tcb3, each of which are exclusively localized to the cER, are encoded by messenger RNAs (mRNAs) that are transported into the growing bud by the myosin motor protein Myo4, the mRNA-binding protein She2, and the Myo4 adaptor protein She3 (Shepard et al., 2003). Myo4 and She3 are also required for the efficient inheritance of cER, which is derived from nuclear ER and is carried by Myo4 into the bud along actin cables (Estrada et al., 2003). Thus, Myo4 transports ER into the bud and delivers mRNAs that encode ER-PM tethering proteins for cortical ER formation. This system likely evolved because pre-existing cER membrane proteins in the mother cell, such as Ist2 and the tricalbins that localize exclusively to the cER, are unable to enter the bud due to the proposed diffusion barrier created by the septin ring at the mother-bud neck. In contrast, Scs2 localizes to nuclear and cytoplasmic ER structures in addition to the cER, and thus pre-existing Scs2 protein can be inherited from the mother cell. Cells lacking Scs2 have cER defects that are more pronounced in the bud (Loewen et al., 2007). Thus, Scs2 may initially establish the cER in the newly forming bud. Subsequently, Ist2 and the tricalbins further stabilize cER-PM contacts following their translation and insertion into the newly formed cER.

Roles for PM-ER Junctions

The identification of proteins that function in the establishment of PM-ER junctions provides a foundation for future investigations into the roles for ER-PM contacts. We found that loss of PM-ER connections resulted in the dramatic accumulation of PI4P at the PM, consistent with previous work indicating that Sac1 functions at PM-ER junctions (Stefan et al., 2011). PI 4-kinase activity has been implicated in several important cellular processes at the PM, including signaling responses to numerous stimuli in both yeast and metazoan cells (Audhya and Emr, 2002; Balla et al., 2008; Jesch et al., 2010; Hammond et al., 2012; Yavari et al., 2010). ER-PM contacts serve as important sites for the modulation of PI4P levels and may contribute to the regulation of cell signaling pathways controlled by PI4P. It will also be interesting to determine whether processes regulated by PM-ER junctions in metazoan cells (e.g., lipid and Ca^{2+} transport) are affected in Δtether yeast cells.

Unexpectedly, cells lacking cER exhibited continuous UPR signaling and required the UPR for viability, suggesting one or more critical processes in the ER require PM-ER contacts for full function. In the absence of PM-ER connections, these processes are perturbed, resulting in imbalances in the ER and induction of the UPR. Thus, constitutive UPR signaling may be necessary as a protective mechanism to compensate for the disruption of ER homeostasis upon loss of PM-ER tethers. The compromised ER function in the Δtether cells and synthetic growth defect of Δtether cells lacking *IRE1* was not due to impaired protein trafficking out of the ER, as biosynthetic sorting and processing of the cargo protein CPY were normal. In addition, even with strong constitutive UPR signaling in Δtether cells, expression levels of the UPRE-LacZ reporter and the ER chaperone Kar2 (BIP) were further induced when Δtether cells were treated with a protein folding stress. These results suggest that the constitutive UPR signaling in Δtether cells is not the accumulation of unfolded proteins, but misregulation of additional processes in the ER. Consistent with this, cells impaired in several ER processes (e.g., protein glycosylation and folding; the ERAD pathway, ion transport, and lipid metabolism) exhibit elevated UPR signaling (Jonikas et al., 2009) as well as synthetic growth defects upon inactivation of the UPR (Thibault et al., 2011).

Altogether, our study has (1) revealed a complex system composed of multiple tethering factors involved in the formation of PM-ER membrane junctions, (2) provided a more detailed understanding of phosphoinositide signaling at the PM, and (3) uncovered an unexpected role for ER-PM contacts in ER organization and function. In addition, the ER tether proteins possess lipid-binding domains and may serve as sensors of membrane composition to transmit signals between the ER and PM. Upon loss of PM-ER signaling, cells may become dependent on other cell signaling and stress response systems, such as the UPR, to maintain organelle homeostasis and integrity.

EXPERIMENTAL PROCEDURES

Yeast Strains, Plasmids, and Growth Assays

All strains and plasmids used in this study are listed in the Supplemental Experimental Procedures. For all plating assays, cells were grown to midlog, adjusted to 1 OD_{600}/ml, and serial dilutions were plated on the indicated media. For liquid growth assays, cells were grown to midlog, adjusted to 0.1 OD_{600}/ml, and grown at 26°C. OD_{600} measurements were taken at the indicated time points.

Fluorescence Microscopy

All fluorescent microscopy was performed on midlog cells. Images were acquired with a CSU-X spinning disk microscope (Yokogawa) with a 63× 1.4NA objective on an inverted microscope (DM16000B; Leica), a QuantEM EMCCD camera (Photometrics), and controlled by Slidebook 5.0 (Intelligent Imaging Innovations). The brightness and contrast of images were adjusted in Slidebook 5.0 and cropped in Photoshop (Adobe).

Electron Microscopy

Yeast cells were fixed with 2.5% (v/v) glutaraldehyde for 1 hr and incubated in 1% potassium permanganate for 1 hr. Cells were dehydrated with 50%, 70%, 95%, and 100% ethanol. Samples were transitioned into 100% propylene oxide and embedded in Spurr's resin. Electron microscopy was performed on a Morgnani 268 transmission electron microscope (FEI) with an AMT digital camera and 80 eV beam. The cER/PM ratios were measured using ImageJ. The brightness of EM images was adjusted with Photoshop (Adobe).

Phosphoinositide Analysis

Cellular phosphoinositide levels were quantified as described (Foti et al., 2001) with modifications, as described in the Supplemental Experimental Procedures.

Subcellular Fractionation

Cell lysates were loaded on top of 10 ml 20%–60% linear sucrose gradients. Gradients were spun at 100,000 × g for 18 hr in a SW41TI rotor (Beckman). Fractions were taken from the top and analyzed by immunoblotting. See Supplemental Experimental Procedures for more details.

β-Galactosidase Assays

Strains harboring a UPRE reporter (Cox and Walter, 1996) were grown to mid-log, and indicated samples were treated with 8 mM DTT for 1 hr. Cells were harvested and β-galactosidase activity was measured as described in the Supplemental Experimental Procedures.

SUPPLEMENTAL INFORMATION

Supplemental Information includes seven figures, four tables, and Supplemental Experimental Procedures and can be found with this article online at http://dx.doi.org/10.1016/j.devcel.2012.11.004.

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